

Chapter 5

Pain: why and how does it hurt?

by LORIMER MOSELEY

...then, [Mr Hammerhead Shark] his shirt covered in blood, spun around and hit his knee on the table, at which point he swore and yelled 'My knee! My knee!', the whole time unfussed about the hammer stuck in his neck.

G Lorimer Moseley, *Painful Yarns. Metaphors and stories to help understand the biology of pain*

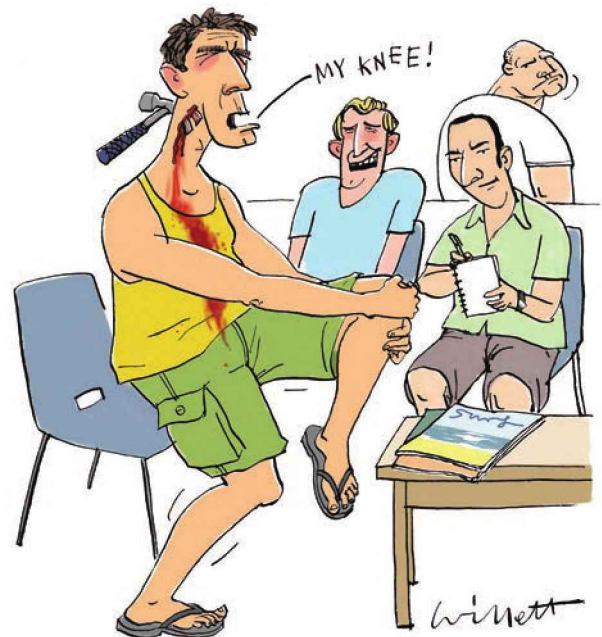
Even the simplest biological organisms can protect themselves from threatening stimuli—by altering their path of movement away from the source of the threat.¹ As evolution has honed us into more and more sophisticated creatures, we have also honed this fundamental capacity to protect ourselves from threat. Indeed, perhaps our most sophisticated protective strategy is pain. In this chapter, some of the 'fearful and wonderful complexity' of pain is conveyed, by:

- outlining a contemporary definition of pain that is contrary to conventional definitions but which integrates the huge amount of research that has been undertaken since our conventional definitions were established
- introducing the idea of nociception and describing some of what is known about the biological mechanisms that underpin nociception
- providing a conceptual framework with which to make sense of pain within the context of clinical practice.

WHAT IS PAIN?

Almost everyone experiences pain. Those who do not experience pain, as the rest of us know it, are at a distinct disadvantage in life and are likely to die young without living fast. Pain is an unpleasant sensory and emotional experience that is felt in the body and motivates us to do something to escape it. These two characteristics of pain—its unpleasantness and its anatomical focus—are what makes it such an effective protector.

Pain alerts us to tissue damage or the threat thereof. Pain makes us seek care. Pain stops us competing, keeps us seeking a cure, compels us to prioritise pain relief above almost everything else. That's the rub—pain changes our behaviour. In fact, if the brain concludes that there is



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something more important than protecting a body part then it makes the executive decision to *not* produce pain. Therein lies the key to really understanding pain; it is as simple and as difficult as this—if *the brain* concludes that a body part is in danger and needs protecting and you, the organism, ought to know about it, then *the brain* will make that body part hurt.²

There is a critical caveat here, one that was overlooked in the fourth edition of this book. This convention to

ascribe pain to the brain is flawed, because a brain, on its own, almost certainly would not and probably could not, produce pain. This position has been criticised, with merit and additional pieces have been authored that critique it.³ The caveat, then, is this: to attribute it all to the brain is simplistic and denies the role of physiological mechanisms that extend beyond the brain. However, as much as we know now, networks of brain cells (neuronal and non-neuronal) are the most obvious 'last step' and the emergence of consciousness (and therefore pain) is likely to be most closely related to their activation.

With that rather perplexing caveat out of the way, let's consider what is *good* about this concept of pain. This concept of pain integrates a vast body of basic, applied and clinical research. It differs greatly from conventional theories, which have changed little since the 17th century when René Descartes was ridiculed for suggesting that we were not made from four bodily humours.⁴

PRACTICE PEARL

The critical concept is that pain is not a measure of tissue damage, but an indicator of the brain's conviction about the need to protect certain tissue.

To better understand pain as the protective output of the brain, not as a marker of tissue damage, let us consider several contrasts between the two models (Table 5.1).

WHAT IS NOCICEPTION? CLUE— NOCICEPTION IS NOT PAIN!

At the risk of sounding a bit repetitive, let us start by saying that nociception is not pain. Nociception refers to the

detection, transmission and processing of noxious stimuli. A noxious stimulus is one that is actually or potentially damaging. The neurons that detect noxious stimuli and transmit a nociceptive message to the spinal cord are called nociceptors ('danger receptors').

Nociceptors are high threshold neurons, which means that they respond to stimuli that are approaching or surpassing that which is damaging to the tissue in which the neuron resides. Nociceptors are mostly thin neurons and many are not myelinated. They fall into three nerve fibre types: Aβ (fast, myelinated neurons), Aδ (slow, myelinated neurons), C (very slow, unmyelinated neurons; see ⁵ for full review).

Nociceptors are located in almost all the tissues of the body, with the notable exception of the brain. This network of neurons, can be considered a very thorough surveillance system. Of course, the surveillance function of the peripheral nervous system is much more comprehensive than nociceptors alone—it is just that nociceptors are always surveying the anatomical landscape for dangerous events. All such events fall into one or more of three categories: thermal, chemical or mechanical. Thus, nociceptors have specialised receptors in their cell walls that are cold-sensitive, hot-sensitive, chemosensitive or mechanosensitive.

Humans have low-threshold neurons that are solely interested in one modality or another, such as thermo-sensitive Aβ fibres, which inform brainstem areas of even tiny fluctuations in tissue temperature—fluctuations that are well within a safe operating range. In contrast, nociceptors are most often bimodal or multimodal. That is, they are responsive to thermal and mechanical input, or to thermal, mechanical and chemical input. These nociceptors, which are situated in the tissues of

Table 5.1 Contrasts between (a) pain as a protective output of the brain and (b) pain as a marker of tissue damage

A Pain as a protective output of the brain	B Pain as a marker of tissue damage
Pain is in consciousness.	Damage is in the body.
One cannot be in pain and not know about it.	One can be severely damaged and not know about it.
No brain, no pain.	No body, no damage.
Pain is affected by what else is at stake.	Damage is not affected by what else is at stake.
Pain is affected by who is in the area.	Damage is not affected by who is in the area.
Pain is affected by beliefs.	Damage is not (well, not directly).
Pain can occur in a body part that does not exist.	Damage cannot occur in a body part that does not exist.
Pain can occur in a body part that is not damaged.	Damage can occur in a body part that is not painful.
Pain can occur without activation of nociceptors (see next section).	Damage cannot (excepting local anaesthetic or pre-injury nociceptor death).

the body, are called primary nociceptors (see ^{6, 7} for review).

Primary nociceptors are also different to other peripheral neurons in that they project to neurons in the dorsal horn of the spinal cord, not directly to thalamic or cortical structures. The neurons with which primary nociceptors synapse in the spinal cord are called secondary or spinal nociceptors and the synapse is open to modulatory input from other peripheral inputs and from descending neurons (see below). That the nociceptive system is polymodal and has a 'relay station' in the spinal cord raises two very important issues.

That primary nociceptors are multimodal and only project as far as the spinal cord clearly shows that the nociceptive system, per se, is not able to transmit information specific to each modality. That is, the nociceptive system does not tell the brain that something is 'dangerously hot', or 'dangerously cold' or 'dangerously squashed'. Rather, the nociceptive system has the apparently simple task of telling the brain that something is 'dangerous'. It is the non-nociceptive inputs (including non-somatosensory cues⁸) that provide critical information about the nature of the danger.

The polymodal characteristic also means that if a stimulus is both dangerously hot and dangerously squashing, it evokes quicker firing of primary nociceptors, which effectively tells the spinal nociceptor that something is 'doubly dangerous'. To consider a clinical example, if a primary nociceptor is activated by chemicals released by an inflammatory event and the tissues are then poked and prodded, the addition of a mechanical input to the chemical input will increase firing of nociceptors to a greater extent than either would alone.

That primary nociceptive input is open to modulation at the spinal 'relay station' (better conceptualised as a spinal 'processing station') means that other peripheral input can decrease noxious input. Peripheral input at the spinal relay station is from interneurons that are activated by wide diameter peripheral neurons (A β fibres) from the same or adjacent area. This is why one can, for example, 'rub it better', or, in a more sophisticated way, put TENS on it. In fact, TENS was born from Melzack and Wall's famous gate control theory of 1965.⁹ Moreover, the spinal relay station can be modulated by descending input from supraspinal structures and it is this descending input that arguably represents a more important and potent modulatory influence.¹⁰

Sensitisation of primary nociceptors ('peripheral sensitisation')

Primary nociceptors become sensitised in the presence of chemical irritants, that is, they are subsequently activated at a lower threshold than is ordinarily required. The most common chemical irritants are generated by the tissues

themselves when they are injured. These injured tissues are often referred to as 'inflammatory soup' because there are many chemicals involved and the exact ingredients of any particular soup will vary, depending upon the particular situation/injury/context/chemical stimuli and, critically, person.

Figure 5.1 depicts typical inflammation-mediated sensitisation of primary nociceptors. This 'peripheral sensitisation' is exactly that—nociceptors become responsive

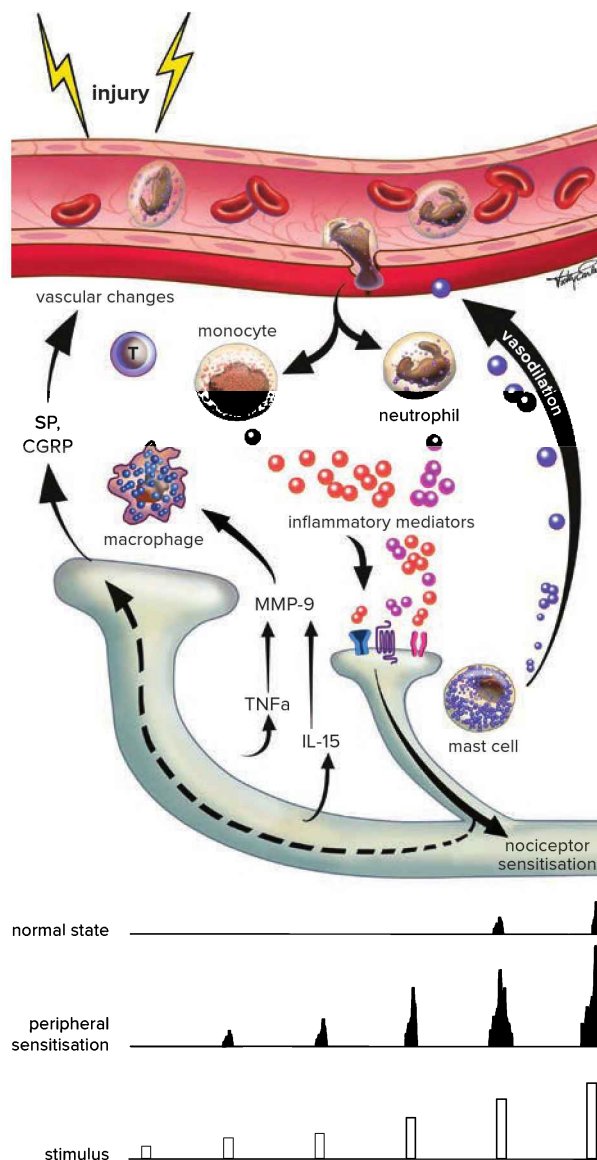


Figure 5.1 A simplified illustration of the connection between nerve terminals and vasculature. The green (normal) and red (sensitised) bars depict nociceptor responses to test stimuli (yellow bars). When there is peripheral sensitisation (red bars), even tiny stimuli evoke responses

to stimuli that are not normally evocative. One obvious example is that of sunburn: sunburnt skin hurts when you get into a shower of 40°C because nociceptors are sensitised sufficiently to be activated by a thermal stimulus 4–5°C cooler than that which would normally be required. That peripherally sensitised tissues are heat-sensitive is a very important phenomenon for the reasoning clinician because sensitisation of the spinal nociceptor does not result in heat sensitivity (Fig. 5.2). Therefore, if one has concluded that there is a peripheral problem and one can heat the culprit tissues to 42°C, one can confirm the conclusion or question it, by determining whether the tissues are more sensitive in the presence of thermal stimuli that *normally* do not activate nociceptors (Fig. 5.2).

Another aspect of sunburn, the reddening of the skin, is an important aspect of peripheral sensitisation. Reddening of the skin is a sign of neurogenic inflammation (Figs 5.1, 5.2). When nociceptors are activated, an impulse is transmitted along every branch of the nociceptor.

If an impulse transmits 'in the wrong direction' and arrives at another terminal branch, then it causes the release of chemicals that in themselves are inflammatory and cause vasodilation.⁵ This mechanism is responsible for the flare one gets around a skin wound or scratch. It is an important mechanism if the nociceptor is being activated proximally, for example, in the dorsal root ganglion or in the spinal cord, because it means that the tissues become inflamed as a consequence of action potentials being propagated elsewhere—in this situation, the tissues are not generating the inflammation.

Sensitisation of spinal nociceptors ('central sensitisation')

When spinal nociceptors are active for some time, they too become sensitised. Many biological processes that can contribute to 'central sensitisation' have been uncovered, and while in-depth discussion is beyond the scope of this chapter, there are several resources that discuss central

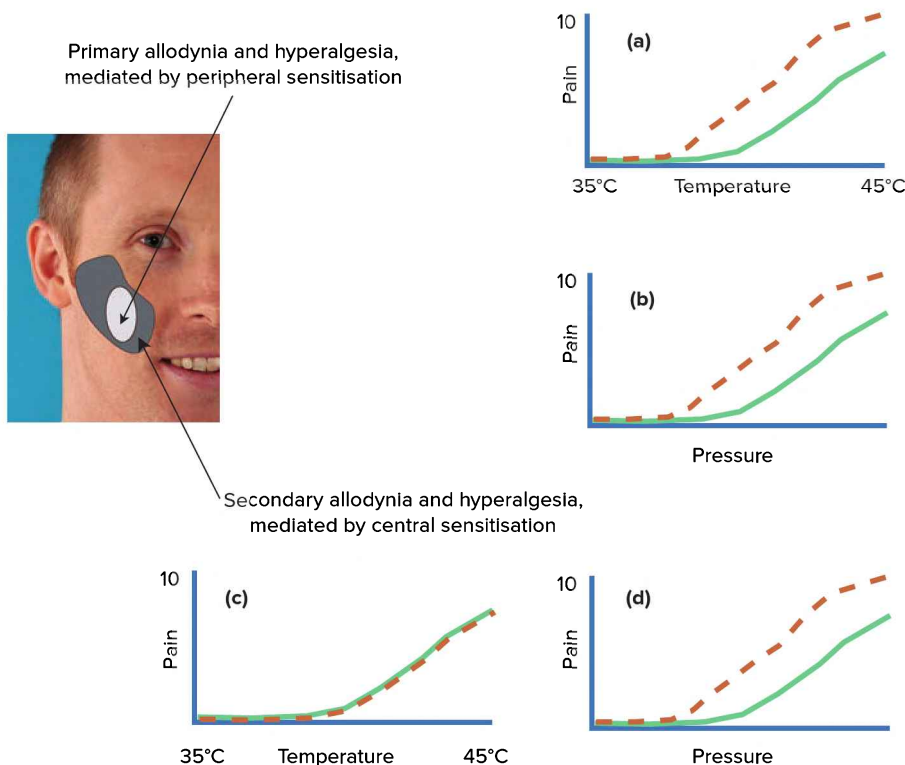


Figure 5.2 Peripheral and central sensitisation.

Presuming an injury within the lightly shaded zone, the presence of inflammation in the area and activation of primary nociceptors will lead to peripheral sensitisation. This will manifest as primary allodynia and hyperalgesia, as represented by a shift in the thermal pain threshold so that pain is evoked at lower temperatures than normal (a) and a similar shift in the mechanical pain threshold so that pain is evoked at lower pressures than normal (b). If central sensitisation ensues, the surrounding area, here represented by the dark shaded zone, will not be heat sensitive (c) but will be mechanically sensitive (d).

A note on the evolution of meaning of central sensitisation

There has been a shift in the common understanding of central sensitisation, from that presented above, to that presented below. That is, many people with persistent pain have reduced pain thresholds across half or all of their body. The most obvious biological substrate for this sensitivity is the central nervous system. As a result, the term central sensitisation has been commandeered to explain this phenomenon.¹³ This is understandable but may

also be problematic because it takes us away from likely candidate adaptations or problems, to a very generic and exclusion-based approach. One can describe 'new central sensitisation' as 'reduced pain thresholds not explainable by peripheral sensitisation'. In this chapter, the former definition of the term is used because it describes a series of biological adaptations that are well understood at a molecular level.

A

sensitisation in more detail and are listed at the end of the chapter. Central sensitisation manifests as mechanical sensitivity beyond the area of injury and peripheral sensitisation (Fig. 5.2). In short, central sensitisation means that the spinal cord 'upregulates' nociceptive input at the spinal cord (see¹⁰⁻¹² for reviews). This has implications for the biopsychosocial model of pain that is illustrated in Chapter 29.

THE BRAIN DECIDES

As illustrated earlier, pain emerges from the brain and reflects the brain's evaluation of threat to body tissue and the subsequent need for action. As mentioned in the earlier caveat, it is true to say that this perspective is 'neurocentric'³ insofar as pain does not emerge from a nervous system, sitting in isolation from the rest of our biology. The truth is, of course, that pain emerges into consciousness from the entire person. This acknowledgement, however, is perhaps more philosophical than a sport and exercise medicine textbook permits. To iterate, we can avoid this difficulty by conceding that although it is not just a brain that evaluates this input and creates an experience, to conceptualise it this way is faithful to much of our understanding and is far more applicable than a more precise but philosophically challenging conceptualisation.

Spinal nociceptors are important informants about danger to body tissue, but, ultimately, nociception is neither sufficient nor necessary for pain (see Table 5.1). Modern conceptual models of pain highlight this critical role of the brain. One framework that makes this clear is that of cortical representations (see⁷ for a clinician-friendly review). According to this framework, an individual will experience, for example, ankle pain when a network of brain cells, distributed across the brain, is activated. That network of brain cells, then, is considered the neural representation of that individual's ankle pain. As it is a unique and distributed network, one might call it a *neurotag*.⁷ Each of the brain cells that constitute this ankle pain neurotag also contribute to other neurotags and activation or otherwise, of this ankle pain neurotag is

open to modulation at every synapse of every constituent brain cell. This brings an enormous complexity to the neurobiology of pain, although pain is simply one expression of the neurobiology of consciousness, which no one fully understands (see above). For a more in-depth review of this neurotag-based concept as it applies to performance, sports medicine and rehabilitation (see Wallwork et al).^{14, 15}

This conceptual framework means that anything that is represented by that individual's brain *and* which provides credible evidence about the danger to which the ankle is currently exposed, should modulate activity of the 'ankle pain neurotag'. In more clinical terms, anything that provides credible information about the likely danger level should modulate ankle pain. Credible information may relate to likely consequences of damage—ankle damage is more dangerous to a top-level dancer than it is to an archer, a fact that will upregulate the ankle pain neurotag in the dancer. However, a top-level dancer might have the long-held belief that his/her body is indestructible, which would downregulate the ankle pain neurotag; a top-level archer may not have this belief. As you can see, it is a very complex system.

Credible information might relate to other sensory cues—damaging one's ankle on the bend in the final of the 200 m in front of a screaming crowd may very well lead to those same cues upregulating the ankle pain neurotag when those sensory cues occur again. Clearly, this neural complexity has important implications for rehabilitation; the neuroscience of pain and protection suggests that rehabilitation of this individual with ankle pain is not complete until the athlete has been exposed to every credible 'byte' of information that implies danger to the ankle.^{14, 15}

Hopefully the reader can now appreciate why Descartes' idea, that we have pain receptors in the tissues and pain signals are transmitted to the brain,⁴ is as inadequate as it is popular. While its simplicity may be seductive, the ever-increasing body of evidence suggests it simply does not hold up. If, as a clinician or patient, one is to accept the true complexity of pain as evidenced by a huge

amount of experimental and clinical literature (see¹⁶ for review) and to conceptualise pain by the representation framework, then one must also accept that activity in spinal nociceptors is one of many contributors to pain. The truly modern day clinician should be open to non-tissue contributions (i.e. differential modulation exerted by other systems, for example, immune upregulation, central sensitisation and central downregulation) and be alert to evidence of their influence.

Charles Darwin suggested that young scientists should write down the results that do not support their current beliefs because they are the results that are most forgettable.¹⁷ As clinicians we should do the same (see also¹⁸). With regard to pain, this might mean taking note of:

- when the same mechanical input flares a condition one day and not the next
- when pain is worse in competition than it is in training
- when strength, endurance, control and flexibility are exemplary but the athlete still 'tweaks' a hamstring running at 90% effort
- when a 20-second manual therapy technique increases hip range of motion by 25°.

Sure, one might squeeze some such findings into a Descartian framework, but they arguably fit more easily in a modern knowledge-based framework and Ockham's razor* compels us to look there first.

THE BRAIN CORRECTS THE SPINAL CORD

There are several hundred projections from the brain to the spinal nociceptor. Projections originate throughout the brain and have both facilitatory and inhibitory effects. Structures within the brainstem, for example, the periaqueductal grey (PAG) are important relay stations between the brain and the spinal nociceptor.¹⁹ Such a powerful modulatory capacity evokes, not surprisingly, powerful modulation.

Many experiments have investigated the circumstances under which the brain modulates the spinal nociceptors. We can easily summarise them here using the same language as above: if the *entire package* of credible evidence of danger is different from that suggested by spinal nociceptor input, the brain will 'correct' the spinal cord by either inhibiting spinal nociceptors or facilitating them.

Those familiar with principles of motor control will recognise the same principles also in the pain system. In motor control, this idea of re-afference and sensory-motor feedback loops is well established—a motor command generates an efferent copy that is then compared to sensory

feedback of the movement (from vision and proprioceptors) and any discrepancy between the predicted and actual outcome is used to correct the motor command.²⁰

Going back to our example of ankle pain, we can describe it thus: activation of the brain's ankle pain neurotag also triggers its efferent copy, the latter of which is compared to spinal nociceptive input, triggering correcting descending modulation to adjust ascending 'danger' input (Fig. 5.3). This kind of feedback loop is common in human physiology.

PRACTICE PEARL

In short, if the brain concludes that the need to protect tissues is greater than spinal nociceptor activity would suggest, it will facilitate the spinal nociceptor. If the brain concludes the need to protect is less than spinal nociceptor activity would suggest, it will inhibit it.

The effects of such correction will be different if the spinal cord itself is sensitised. That is, if non-tissue factors are increasing the brain's evaluation of danger to a particular tissue, such that the brain upregulates the spinal cord, it has the capacity to maintain or indeed instigate, central sensitisation. There is potential here for a vicious cycle of increasing nociception → increased perceived danger → increased descending facilitation → increased nociception. Thus, it is all the more important for the clinician to evaluate *every* factor that is contributing to the perception of threat to tissues because *any* credible evidence of danger has the potential to upregulate the spinal nociceptor (see^{18, 21} for reviews).

THE BRAIN IS DIFFERENT IN THOSE WITH PERSISTENT PAIN

In much the same way that spinal nociceptors adapt to become more sensitive, so too do the brain cells that underpin pain²²⁻²⁵ (Wand et al.²⁶ have reviewed the cortical changes in people with back pain). That is, the more the pain neurotag is active, the better it gets at being active. This manifests in more and more advanced hyperalgesia and allodynia, extended across modalities and exhibiting 'over-generalisation'.

Over-generalisation refers to that phenomenon in which pain begins to be evoked by more innocuous stimuli, in different contexts and under different circumstances.²⁷⁻²⁹ Spreading pain, unpredictable pain, and pain that is less

*Ockham's razor is attributed to William of Ockham (14th century), although it was thought to be 'invented' well before his time. Its earliest and most enduring iteration is 'Entities must not be multiplied beyond necessity' or far more commonly now 'The simplest answer is probably the correct one'.

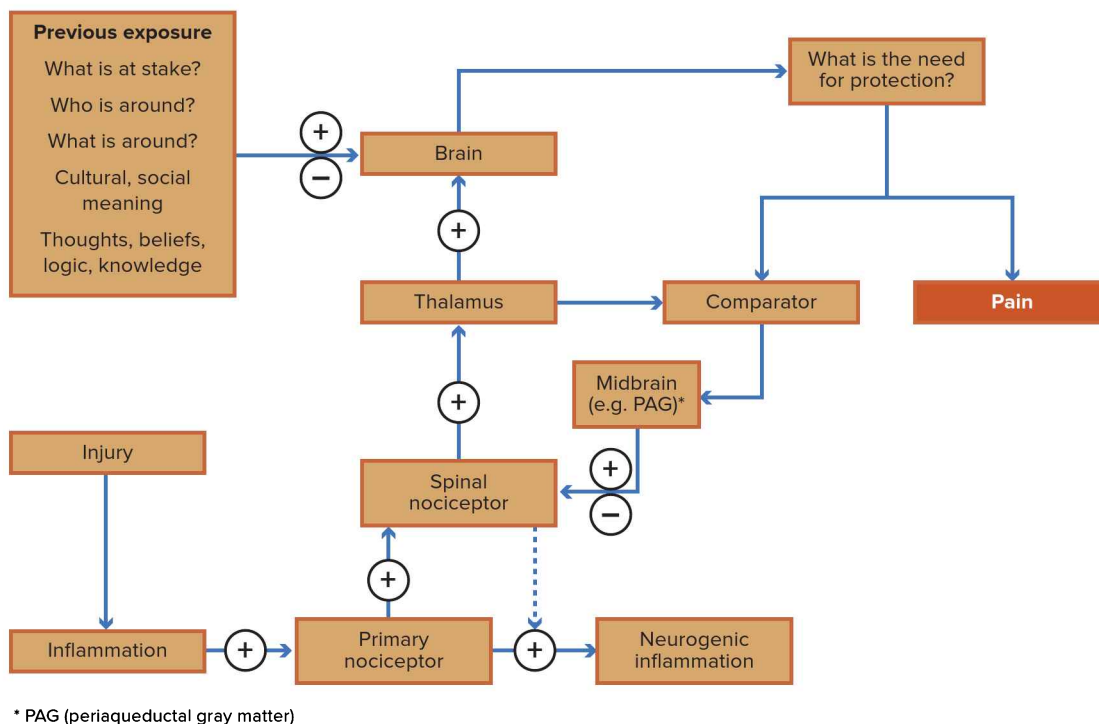


Figure 5.3 Feedback loops within the nociceptive system and the endpoint of pain: injury excites primary nociceptors. Injury-induced inflammation activates and sensitises primary nociceptors (peripheral sensitisation).

Activation of nociceptors causes neurogenic inflammation in nearby areas and excites spinal nociceptors. Spinal nociceptors project to thalamic nuclei, which then project to the brain. A complex evaluative process occurs within the brain, whereby every byte of information that provides credible evidence about the actual danger faced by the tissues in question is able to upregulate or downregulate the pain neurotag. This process permits the brain to determine 'What is the need for protection?' The final 'decision' leads to activation of the pain neurotag and pain emerges into consciousness. Simultaneously, as though a bifurcation of a single neural output, the determined need for protection is sent to a 'comparator', where it is used as a reference for the spinal nociceptor input. This comparator then modulates midbrain structures and thence in turn the spinal nociceptor to 'correct' its activation level. Thus, descending modulation can be facilitatory or inhibitory. The broken line from spinal nociceptor towards neurogenic inflammation refers to the possibility of tissue inflammation evoked by descending facilitation. That is, excessive facilitation of the spinal nociceptor in the absence of primary nociceptor activity is thought to activate the primary nociceptor antidromically, which induces neurogenic inflammation in the periphery.

and less related to tissue state and activity are cardinal signs of cortical dysfunction.

The full mechanisms and manifestations of cortical sensitivity are not as well understood as the changes in the spinal nociceptor are understood, but they are, at least theoretically, likely to be of far more widespread impact for the clinician and patient. Suffice it to state here that the complexity of pain and the adaptability of the nervous system mean that the modern clinician needs to think well beyond the tissues when dealing with anyone in pain. Indeed, common changes in the sensitivity of the nociception/pain system can be mediated at various levels of the neuraxis (Table 5.2).

TREATING SOMEONE IN PAIN: A COMPLEX SYSTEM REQUIRES A COMPREHENSIVE APPROACH

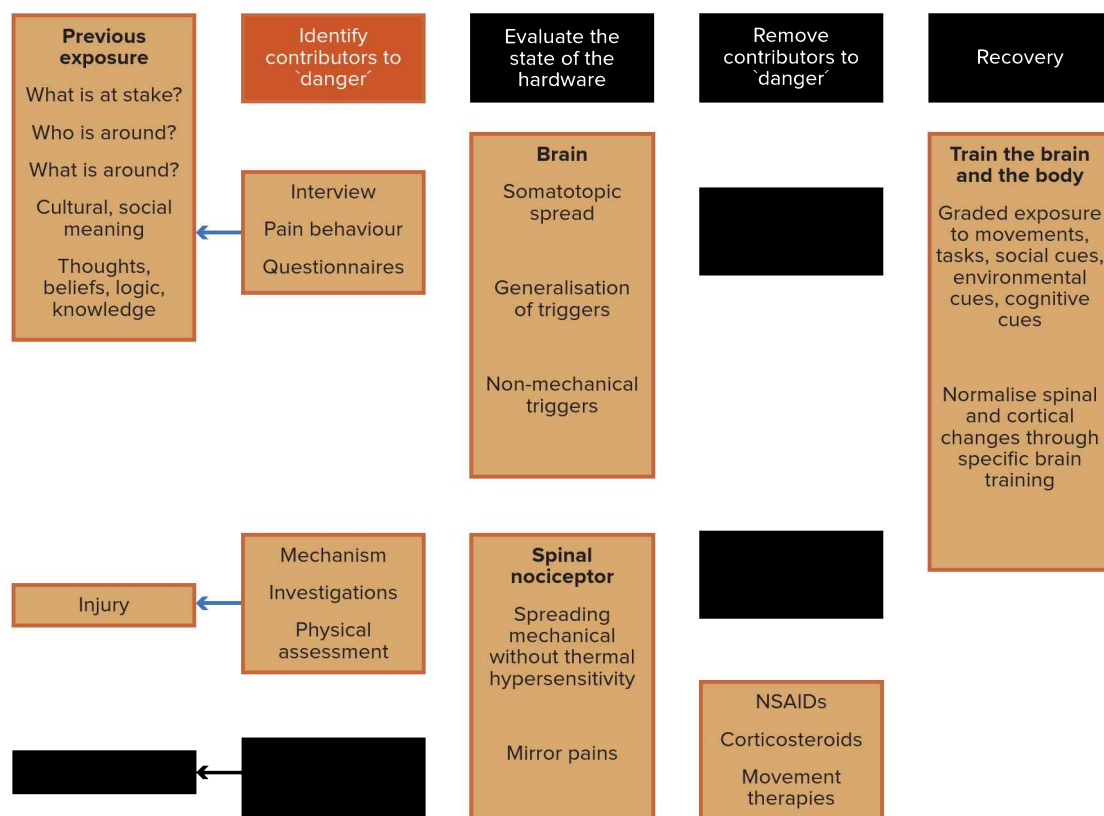
This chapter is not designed to provide a comprehensive guide to treating the patient in pain. Instead, it provides an account of pain that can underpin the assessment and management of people in pain, whatever the 'clinical toolbox' of the clinician. One implication of what we now know about pain is that there are certainly 'many ways to skin a cat' insofar as there are many ways to decrease credible evidence of danger and increase credible evidence of safety (see Moseley and Butler²¹ for a step-by-step approach to doing this).

Table 5.2 Clinical patterns of increased sensitivity to peripheral stimuli and possible underlying mechanisms

Clinical manifestation	Possible underlying cause
Mechanical allodynia: mechanical stimuli that do not normally evoke pain now do	Peripheral sensitisation, central sensitisation, cortical modulation
Thermal allodynia: heat pain threshold is decreased	Peripheral sensitisation, cortical modulation
Hyperalgesia: normally painful stimuli are now more painful	Peripheral sensitisation, central sensitisation, cortical modulation
Primary hyperalgesia	Hyperalgesia attributed to peripheral sensitisation
Secondary hyperalgesia	Hyperalgesia attributed to central sensitisation (or simply not attributable to peripheral sensitisation)

The best practice approach to rehabilitation integrates the complexity of the individual, and targets aspects of normal and altered physiology associated with pain (Fig. 5.4). The model is most established for the

management of people in chronic pain² but can also be applied to the sporting and performance context.^{14, 15} This is extremely relevant to sport and exercise medicine because (i) active people often present with these

**Figure 5.4** Major contributions to pain and their clinical implications.

The left column lists the major contributions to the brain's evaluation of danger to tissues, which determines pain and descending modulation. The second column suggests mechanisms with which to assess the major contributions, thereby identifying key triggers. The middle column suggests signs of spinal and cortical sensitisation, so as to determine the state of the nociceptive system. The fourth column suggests approaches to minimise the impact of the triggers that have been identified. (Note that this extends beyond physical approaches to include cognitive and educational approaches.) The final column recommends avenues to recovery. (Note the emphasis on training both the body and the brain, via specific techniques and graded exposure to physical and non-physical triggers.)

symptoms and (ii) the skillset and credibility of those working in this field is well suited to motivating people in pain to embrace a multifactorial approach to management. Because the biological mechanisms that underpin pain are the same for acute and chronic pain, albeit with increasing

sensitivity as pain persists, the model outlined is applicable across patient groups. In Chapter 6, Dr Ebonie Rio outlines approaches to several clinical scenarios to show how the principles outlined here can underpin work in the clinic.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

